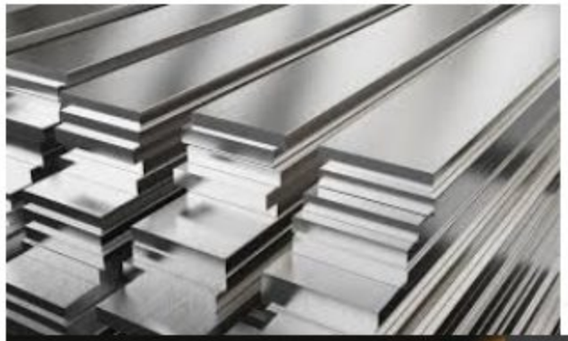
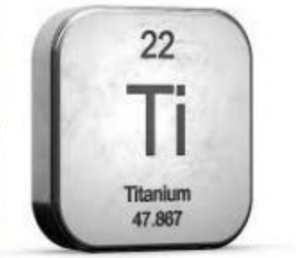
Material Properties

Durability

Durability is a material's ability to withstand wear, pressure, corrosion and damage over a materials lifespan whilst still being able to function. Examples of materials having the property of durability can range from resisting corrosion in the open air to being able to resist scratches and compressive pressure. There’s a lot of materials that have the property of durability like metals such as stainless steel, titanium and bronze, ceramics like porcelain, composites and a select few polymers.

Some materials have very poor durability like plastics and glass due to their inability to resist scratching and the fact that they shatter easily. Another material with poor durability is plain carbon steel as it rapidly rusts when exposed to oxygen, causing it to require coating for it to be used.







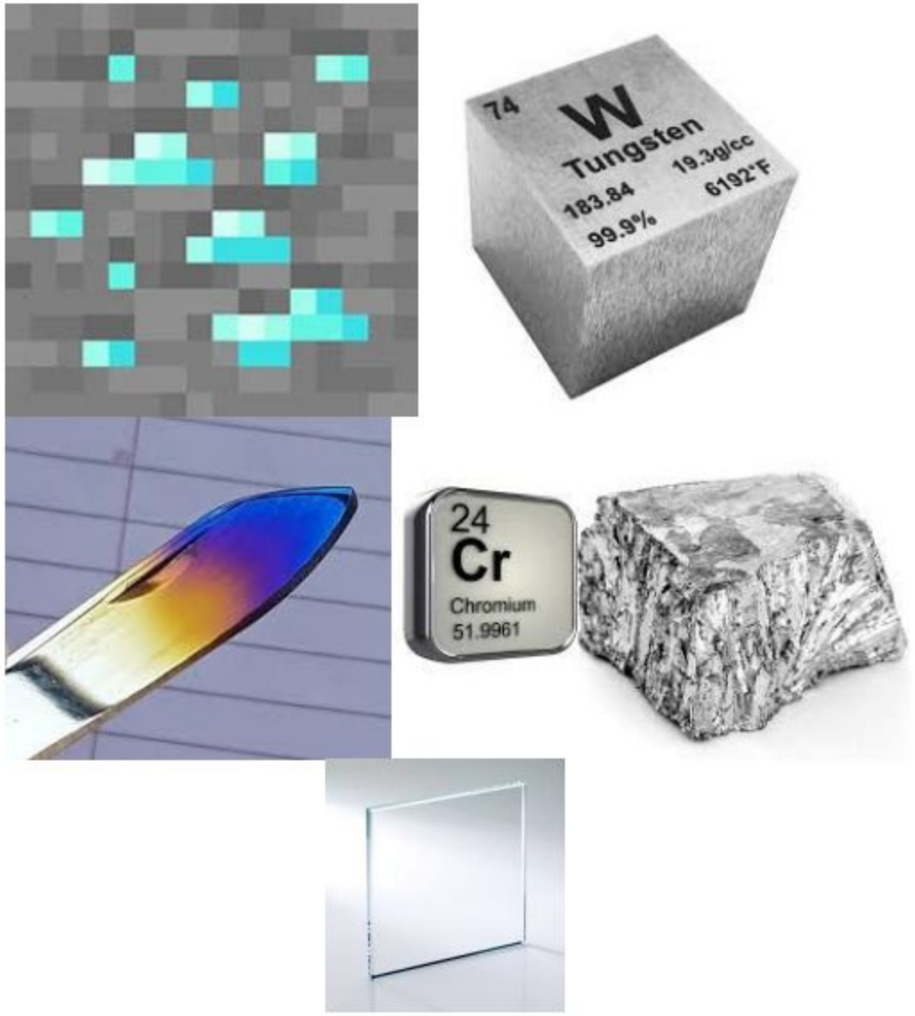




Hardness

Hardness is a material's ability to resist penetrating, scratching and abrasive forces. This property is found in materials used in applications such as gears, cutting instruments such as drills and saws and also in construction. A material cannot be both hard and tough as they are contrasting. Examples of hard materials are glass, diamond (y: -52) as well as metals such as Tungsten, Tempered Steel and Chromium.

When it comes to materials that demonstrate poor hardness there are alot. Metals such as tin and aluminium have poor hardness as they can be easily marked and scratched. Other examples are annealed copper or zinc, which are heat treated to become softer and more malleable.



Toughness

Toughness is a material's ability to absorb energy and deform without fracturing. This energy is normally transferred when the material is impacted or has to carry a sudden load. The difference between strength and toughness is that strength is the resistance of static loads whereas toughness is the resistance of dynamic and impact loads. Examples of materials with high toughness are metals such as carbon steel, graphene and wood like timber.

Materials with poor toughness are brittle materials like glass, porcelain, chalk and concrete. All of these materials have high hardness but when it comes to impact the crack under the pressure.

